CHAPTER FOUR: PRESENTATION OF RESEARCH FINDINGS

This chapter reveals the results of the data collection and analysis. It starts by outlining the response rate and the reliability of the research tool that was utilized. Then, it provides an overview of client sociodemographic characteristics, perceptions of health service quality, and relationships between clients' sociodemographic characteristics and perception of health service quality.

4.1 Reliability and Validity

A pilot study was carried out prior to conducting the primary study. A total of 36 individuals made up a convenient sample in the pilot study: Three academic/professional experts, three senior managers, ten middle-level managers, ten employees and ten clients. Prior to conducting the primary investigation, the questionnaire's reliability was assessed using Cronbach alpha. Table 4.1 presents the results.

Table 4.1. Cronbach alpha test results

Reliability Statistics		
	Cronbach's Alpha	**No. of items**
Patient expectations of service quality	0.916	22
Patient perceptions of service quality	0.952	22

Primary data (2024)

The range of Cronbach's alpha is r = 0 to 1. The data gathering instrument is considered sufficiently reliable when r = 0.7 or higher (Hair et al., 1995). The questionnaire in this study was regarded as reliable because the Cronbach alpha for statements about clients' expectations of service quality and their perception of the same exceeded 0.7. Feedback from academic/professional experts and managers at Halisi Family Hospital confirmed that the questionnaire was valid and measured what it intended.

4.2 Response Rate

Three hundred thirty (330) online questionnaires were sent to 330 clients via WhatsApp. Thirty-four (34) respondents were not on WhatsApp, and 14 did not consent to participate in the survey. This reduced the study's sample size to 282 clients. The number of questionnaires completed was 174. Consequently, a 61% overall rate of response and a 38.3% non-response rate were achieved. Figure 4.1 below shows a visual representation of the rate of response.

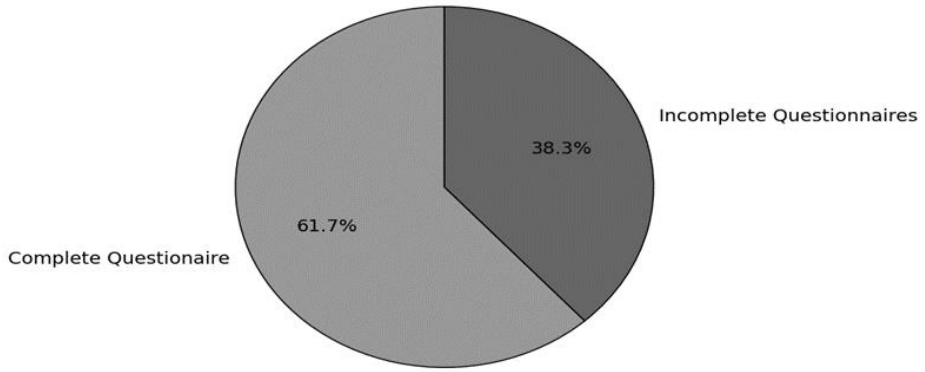


Figure 4.1: Rate of response: Primary data (2024)

4.3 Clients' Sociodemographic Characteristics

This sub-section presents the respondents' sociodemographic characteristics. It covers their gender, age, education level, work status, mode of payment for health care services, and income.

4.3.1 Gender

48.9% of those surveyed were male, 48.3% were female, 2.3% preferred not to reveal their gender, and 0.5% identified as someone of a different gender, according to the data. The number of males and females is relatively equal, indicating that the results will not be gender biased. Table 4.2 provides a summary of the findings.

Table 4.2: Gender

Gender	Frequency	Percentage (%)
Male	85	48.9
Female	84	48.3
Prefer not to say	4	2.3
Other	1	0.5
Total	**174**	**100.0**

Primary data (2024)

4.3.2 Age

63.8% of respondents were between the ages of 18 and 44, 29.9% were between the ages of 45 and 59, 1.1% were older than 60, and 5.2% chose to not reveal their age. Individuals aged 18 to 59 made up the majority of the respondents. Table 4.3 provides a summary of the findings.

Table 4.3: Age

Age	Frequency	Percentage (%)
18 – 44 years	111	63.8
45-59 years	52	29.9
>60 years	2	1.1
Prefer not to say	9	5.2
Total	**174**	**100.0**

Primary data (2024)

4.3.3 Education Level

As per the results of the study, 48.9% of respondents had completed their secondary education, 17.2% stated that they would prefer not to disclose their educational history, 16.7% had a certificate or diploma, 12.1% had a postgraduate degree, and 5.2% had finished their undergraduate studies. An overview of the results is given in Table 4.4 below.

Table 4.4: Education Level

Education Level	Frequency	Percentage (%)
Bachelor's Degree	85	48.9
Prefer not to say	30	17.2
Certificate or Diploma	29	16.7
Postgraduate Degree	21	12.1
Secondary Education	9	5.2
Total	**174**	**100.0**

Primary data (2024)

4.3.4 Work Status

The research findings showed that 58.6% of the respondents were employed, 20.7% were self-employed, 9.2% were unemployed, 2.3% were retired while 9.2 % opted not to disclose their work status. Most of the respondents were either employed or self-employed. Table 4.5 provides a summary of the results.

Table 4.5: Work Status

Work Status	Frequency	Percentage (%)
Employed	102	58.6
Self-Employed	36	20.7
Unemployed	16	9.2
Retired	4	2.3
Prefer not to say	16	9.2
Total	**174**	**100.0**

Primary data (2024)

4.3.5 Payment Method for Healthcare Services

The results show that the most common payment method for healthcare services is through private health insurance provided by the employer, covering 40.2% of the respondents. 12.1% use both NHIF and private health insurance purchased by themselves, 9.2% rely solely on NHIF, 8.1% pay for healthcare out of pocket using cash, Mpesa, credit, or debit cards, 6.9% preferred not to disclose their mode of payment, 5.7% use both NHIF and private health insurance provided by their employer, and 5.2% use private insurance purchased by self. Table 4.6 gives an overview of the results.

Table 4.6: Payment method for Healthcare Services

Mode of Payment	Frequency	Percentage (%)
Private Health Insurance provided by my employer	70	40.2
NHIF Only, Private Health Insurance purchased by self	21	12.1
NHIF Only	16	9.2
Out of Pocket - Cash, Mpesa, Credit or Debit Card	14	8.1
Prefer not to say	12	6.9
NHIF Only, Private Health Insurance provided by my employer	10	5.7
Private Health Insurance purchased by self	9	5.2
Out of Pocket – Cash, Mpesa, Credit or Debit Card, NHIF Only	5	2.9
NHIF Only, Private Health Insurance provided by my employer, Private Health Insurance purchased by self	5	2.9
Out of Pocket - Cash, Mpesa, Credit or Debit Card, Private Health Insurance purchased by self	4	2.3
Out of Pocket - Cash, Mpesa, Credit or Debit Card, Private Health Insurance provided by my employer	3	1.7
Private Health Insurance provided by my employer, Private Health Insurance purchased by self	3	1.7
Private Health Insurance provided by my employer, NHIF Only	2	1.1
Total	**174**	**100.0**

Primary data (2024)

4.3.6 Average Monthly Household Income

The results show that 45.4% of the respondents chose not to disclose their average monthly household income, 21.3% earned Ksh. 50,000 – 99,999, 13.2% earned Ksh. 49,999 and below, 10.9% Ksh. 100,000 – 149,000, and 9.2% of the respondents earned Ksh. 150,000 and above. Of the respondents who chose to answer this part of the questionnaire, the majority earn an average monthly household income above Ksh. 50,000.Table 4.5 gives an overview of the results.

Table 4.7: Average Monthly Household Income

Average Monthly Household Income	Frequency	Percentage (%)
Prefer not to say	79	45.4
50,000 – 99,000	37	21.3
< 49,999	23	13.2
100,000 – 149,000	19	10.9
>150,000	16	9.2
Total	**174**	**100.0**

Primary data (2024)

4.4 Expectations

The mean expectations of the clients of Halisi Family Hospital are summarised in Table 4.8 below.

The mean expectations score for all five dimensions is 4.52. The results show that the assurance dimension has the highest mean score and highest expectations (Mean = 4.55; SD= 0.73), followed by the empathy dimension (Mean = 4.54; SD= 0.75), then the responsiveness dimension (Mean = 4.53; SD= 0.79), reliability dimension (Mean = 4.50; SD= 0.77) and lastly the tangibles dimension (Mean = 4.48; SD= 0.78). In terms of individual items in the tangibles dimension, the item "Modern, up-to-date equipment" (Mean = 4.56) is ranked highest, and the item "Visually appealing materials associated with the service" (Mean = 4.41) is ranked lowest. In the reliability dimension, the items "Dependability in solving clients' problems" and "Maintaining accurate and error-free record" (Mean = 4.53) are ranked highest, and the item "Providing service as promised" (Mean = 4.43) is ranked lowest. In the responsiveness dimension, the item "Provide prompt services" (Mean = 4.63) is ranked highest, and the item "Employees are never too busy to respond to patients' requests" (Mean = 4.48) is ranked lowest. In the assurance dimension, the item "Patients feel safe in all their dealings with the hospital and its employees" (Mean = 4.60) is ranked highest, and the item "Courteous employees" (Mean = 4.51) is ranked lowest. Finally, in the empathy dimension, the item "Hospital employees have the patients' best interests at heart " (Mean = 4.62) is ranked highest, and the item "Operating hours are convenient" (Mean = 4.50) is ranked lowest.

Table 4.8: Means, Standard Deviations, and Rank for Clients' Expectations by Dimension

Dimension	Mean	SD	Rank
Tangibles			
Modern, up-to-date equipment	4.56	0.88	**1**
Clean and visually appealing physical facilities	4.53	0.75	**2**
Well-dressed employees with a neat and professional appearance	4.45	0.73	**3**
Visually appealing materials associated with the service	4.41	0.77	**4**
Mean of Dimension	**4.48**	**0.78**	**-**
Reliability			
Providing service as promised	4.43	0.76	**5**
Dependability in solving patients' problems	4.53	0.74	**1**
Providing services at the promised time	4.51	0.77	**3**
Providing services right the first time	4.49	0.86	**4**
Maintaining accurate and error-free records	4.53	0.69	**2**
Mean of Dimension	**4.50**	**0.77**	**-**
Responsiveness			
Employees tell the patients exactly when services will be performed	4.50	0.77	**3**
Provide prompt services	4.63	0.78	**1**
Employees are always willing to help patients	4.52	0.80	**2**
Employees are never too busy to respond to patients' request	4.48	0.82	**4**
Mean of Dimension	**4.53**	**0.79**	**-**
Assurance			
Behaviour and attitudes of hospital employees instil confidence in patients	4.54	0.71	**3**
Patients feel safe in all their dealings with the hospital and its employees	4.60	0.70	**1**
Courteous employees	4.51	0.78	**4**
Knowledgeable employees able to answer patient's questions	4.57	0.72	**2**
Mean of Dimension	**4.55**	**0.73**	**-**
Empathy			
Individualised attention for patients	4.54	0.75	**2**
Care expressed by employees to patients	4.53	0.69	**3**
Employees have the patients' best interests at heart	4.62	0.69	**1**
Employees understand the specific needs of each patient	4.53	0.73	**4**
Operating hours are convenient	4.50	0.87	**5**
Mean of Dimension	**4.54**	**0.75**	**-**
Mean of all Dimensions	**4.52**	**0.67**	**-**

Primary data (2024)

Table 4.9 below includes the statements with the greatest and lowest expectation scores.

Table 4.9. Highest and Lowest Scores on Expectations Statements

Expectations - Highest Scores	
Provide prompt services - Responsiveness	4.63
Employees have the patients' best interest at heart - Empathy	4.62
Patients feel safe in all their dealings with the hospital and its employees - Assurance	4.60
Knowledgeable employees able to answer patient's questions - Assurance	4.57
Modern, up-to-date equipment - Tangibles	4.56
Expectations - Lowest Scores	
Providing services right the first time - Reliability	4.49
Employees are never too busy to respond to patients' requests - Responsive	4.48
Well-dressed employees with a neat and professional appearance - Tangibles	4.45
Providing service as promised - Reliability	4.43
Visually appealing materials associated with the service - Tangibles	4.41

Primary data (2024)

Clients have the highest expectations for prompt service, employees who have their best interests at heart, safety, knowledgeable staff who can answer their questions and modern equipment. These areas are crucial for enhancing client perception of healthcare service quality and should be prioritised by the hospital. While still important, expectations are relatively lower for aspects like getting the services right the first time, responsiveness, professional appearance, keeping promises, and visually appealing materials. These areas should be addressed but may be considered secondary to the top priority areas.

4.5 Perceptions

The mean perceptions of the clients of Halisi Family Hospital are summarised in Table 4.10 below.
The mean perception score for all five dimensions is 4.21. The results in Table 4.10 show that the responsiveness dimension has the highest mean score (Mean = 4.32; SD= 0.72), followed by the assurance dimension (Mean = 4.21; SD= 0.59), then the empathy dimension with (Mean = 4.18; SD= 0.65), the reliability dimension (Mean = 4.18; SD= 0.77) and finally the tangibles dimension (Mean = 4.16; SD= 0.69). In terms of individual items in the tangibles dimension, the item "Clean and visually appealing physical facilities" (Mean = 4.30) is ranked highest, and the item "Modern, up-to-date equipment" (Mean = 4.03) is ranked lowest. In the reliability dimension, the item "Providing services

right the first time" (Mean = 4,37) is ranked highest, and the item "Providing services at the promised time" (Mean = 3.93) is ranked lowest. In the responsiveness dimension, the item "Employees are never too busy to respond to patients' requests" (Mean = 4.35) is ranked highest, and the item "Employees tell the patients exactly when services will be performed" (Mean = 4.30) is ranked lowest. In the assurance dimension, the item "Behaviour and attitudes of hospital employees instil confidence in patients" (Mean = 4.37) is ranked highest, and the item "Courteous employees" (Mean = 4.10) is ranked lowest. Finally, in the empathy dimension, the item "Operating hours are convenient" (Mean = 4.27) is ranked highest, and the item "Employees understand specific needs for each patient" (Mean = 4.04) is ranked lowest.

Table 4.10: Means, Standard Deviations and Rank for Clients' Perception by Dimension

Dimension	Mean	SD	Rank
Tangibles			
Modern, up-to-date equipment	4.03	0.79	**4**
Clean and visually appealing physical facilities	4.30	0.60	**1**
Well-dressed employees with a neat and professional appearance	4.21	0.63	**2**
Visually appealing materials associated with the service	4.13	0.66	**3**
Mean of Dimension	**4.16**	**0.69**	**-**
Reliability			
Providing service as promised	4.02	0.77	**4**
Dependability in solving patients' problems	4.33	0.69	**2**
Providing services at the promised time	3.93	0.81	**5**
Providing services right the first time	4.37	0.72	**1**
Maintaining accurate and error-free record	4.25	0.65	**3**
Mean of Dimension	**4.18**	**0.73**	**-**
Responsiveness			
Employees tell the patients exactly when services will be performed	4.30	0.65	**4**
Provide prompt services	4.34	0.80	**2**
Employees are always willing to help patients	4.32	0.68	**3**
Employees are never too busy to respond to patients' request	4.35	0.72	**1**
Mean of Dimension	**4.32**	**0.72**	**-**
Assurance			
Behaviour and attitudes of hospital employees instil confidence in patients	4.37	0.60	**1**
Patients feel safe in all their dealings with the hospital and its employees	4.26	0.62	**2**
Courteous employees	4.10	0.57	**4**
Knowledgeable employees able to answer patient's questions	4.12	0.57	**3**
Mean of Dimension	**4.21**	**0.59**	**-**
Empathy			
Individualised attention for patients	4.23	0.62	**3**
Care expressed by employees to patients	4.11	0.64	**4**
Employees have the patients' best interest at heart	4.25	0.60	**1**
Employees understand the specific needs of each patient	4.04	0.65	**5**
Operating hours are convenient	4.27	0.70	**2**
Mean of Dimension	**4.18**	**0.65**	**-**
Mean of all Dimensions	**4.21**	**0.76**	**-**

Primary data (2024)

Table 4.9

Table 4.11 below includes the statements with the greatest and lowest perception scores.

Table 4.11. Highest and Lowest Scores on Perceptions Statements

Perceptions - Highest Scores	
Behaviour and attitudes of hospital employees instil confidence in patients - Assurance	4.37
Providing services right the first time - Reliability	4.37
Employees are never too busy to respond to patients' request - Responsiveness	4.35
Dependability in solving patients' problems - Reliability	4.33
Clean and visually appealing physical facilities - Tangibles	4.30
Perceptions - Lowest Scores	
Courteous employees - Assurance	4.10
Employees understand the specific needs of each patient - Empathy	4.04
Modern, up-to-date equipment - Tangibles	4.03
Providing service as promised - Reliability	4.02
Providing services at the promised time - Reliability	3.93

Primary data (2024)

The results in Table 4.11 show that clients had the highest perceptions regarding employee behaviour and attitude, correctness of services, responsiveness, dependability, and cleanliness of facilities. These areas are strengths of the hospital. The lower scores in courteousness, understanding specific client needs, having modern equipment, keeping promises, and punctuality indicate areas where the hospital can improve to enhance overall client satisfaction.

4.6 Perceived Service Quality (Gap Analysis)

Perceived service quality (PSQ), also known as the gap score (G), was calculated by subtracting expectations from perceptions PSQ (G) = PS – ES. The provision of services that fall short of expectations is indicated by a negative gap score. A positive gap score, on the other hand, indicates that the services are better than anticipated.

The Kolmogorov-Smirnov test was applied to assess the normality of the gap distribution. This test shows that the data follows a normal distribution with a p-value > 0.05, as shown in Table 4.12. A histogram was also used to confirm the normality of the continuous data. As a result, all statistical tests used were parametric.

Table 4.12: Kolmogorov- Smirnov Normality Test Results

Kolmogorov- Smirnov Normality Test		
Dimensions	**Statistic**	**Sig.**
Tangibles	0.374	0.524
Reliability	0.278	0.745
Responsiveness	0.209	0.980
Assurance	0.195	0.990
Empathy	0.158	0.998

Primary data (2024)

With the assumption of normality being met, a paired t-test was applied to see if the client's expectations and perceptions of the five service quality dimensions differed statistically significantly. The results are shown in Table 4.13 below.

Table 4.13: Perceived Service Quality by SERVQUAL dimensions

Quality Dimensions	Mean Perception	Mean Expectation	Mean Gap	Rank	P value
Tangibles	4.16 ± 0.69	4.48 ± 0.78	-0.32	3	0.0281
Reliability	4.18 ± 0.73	4.50 ± 0.77	-0.31	4	0.0007
Responsiveness	4.32 ± 0.72	4.53 ± 0.79	-0.21	5	0.0021
Assurance	4.21 ± 0.59	4.55 ± 0.73	-0.34	2	0.0002
Empathy	4.18 ± 0.65	4.54 ± 0.75	-0.36	1	0.0018
Overall Mean	**4.21± 0.76**	**4.52 ± 0.67**	**-0.31**	**-**	**0.0000**

Primary data (2024)

The quality of healthcare services given was found to be below expectations across all service quality characteristics, as indicated by the negative gap scores for all of them. The dimensions with the largest gap scores were empathy (-0.36), followed by assurance (-0.34),then tangibles (-0.32), and reliability (-0.31). With a gap score of (-0.21), the responsiveness component had the lowest gap, indicating the least amount of variance between expectations and perceptions. Additionally, the total gap score was negative (-0.31).

The paired t-test results showed that in all five dimensions, the differences between expectations and perceptions were statistically significant (p-value < 0.05). The overall gap score (0.31) was also statistically significant.

Table 4.14: Statements that scored the highest and lowest gaps

Highest Gap Scores	
Providing services at the promised time - Reliability	-0.58
Modern, up-to-date equipment - Tangibles	-0.53
Employees understand the specific needs of each patient - Empathy	-0.49
Care expressed by employees to patients - Empathy	-0.42
Courteous employees - Assurance	-0.41
Lowest Gap Scores	
Employees are always willing to help patients - Responsiveness	-0.20
Employees tell the patients exactly when services will be performed - Responsiveness	-0.20
Behaviour and attitudes of hospital employees instil confidence in patients - Assurance	-0.17
Employees are never too busy to respond to patients' requests - Responsiveness	-0.13
Providing services right the first time - Reliability	-0.12

Primary data (2024)

As shown in Table 4.14 above, clients reported the lowest perceived service quality (highest gap score) for the statements providing services at the promised time, having modern, up-to-date equipment, employees understanding the specific needs of each patient, employees expressing care to patients and courteous interactions with employees. The lowest gap scores suggest that while still important, expectations are relatively better met in areas such as employees' willingness to help patients, clear communication by employees of when services will be performed, employees not being too busy to respond to patients' requests and providing services correctly the first time.

Addressing the highest gap areas will help significantly improve clients' perception of service quality. Meanwhile, maintaining the performance in areas with the lowest gaps will help sustain the hospital's strengths.

4.7 Influence of Sociodemographic Factors on Perceived Healthcare Service Quality

In this sub-section, one-way ANOVA was applied to examine the relationships between client sociodemographic characteristics (gender, age, work status, education level, mode of payment for healthcare services and income) and their perceptions of healthcare service quality.

4.7.1 Gender

The results in Table 4.15 demonstrate that gender significantly influences clients' perceptions of service quality across all five dimensions since (p-value < 0.05). For each dimension, males generally have more negative gap scores compared to females, indicating that males perceived a larger gap between expected and actual service quality. In the reliability dimension, the "Other" category in the gender variable has a mean gap score of (1±1.26), suggesting that this group perceived a positive gap. They perceived that the services offered were better than their expectations. However, this group had a very negative score for the empathy dimension (-1.4 ± 1.0).

Table 4.15: Gender and Gap Scores

Gender	Tangible	Reliability	Responsiveness	Assurance	Empathy	Overall
Male	-0.47 ± 0.94	-0.63 ± 0.97	-0.65 ± 0.97	-0.59 ± 0.78	-0.52 ± 0.86	-0.57 ± 0.91
Female	-0.27 ± 1.1	-0.34 ±1.12	-0.33 ± 1.16	-0.34 ±1.08	-0.28 ±1.1	-0.29 ± 1.13
Prefer not to say	-0.38 ± 0.68	-0.6 ± 0.80	-0.56 ± 0.9	-0.12 ± 0.63	-0.3 ± 0.82	-0.4 ± 0.78
Other	-0.5 ± 0.71	1 ± 1.26	-0.25 ±1.38	-0.5 ± 1	-1.4 ± 1.0	-0.32 ± 1.27
F- value	4.900	7.280	5.650	5.417	7.474	24.455
P- value	0.002	0.000	0.001	0.001	0.000	0.000

Primary data (2024)

A high F-value indicates a greater degree of variance between the groups compared to within the groups. The highest F-value is observed in the 'Empathy' dimension (F-value = 7.474). This suggests that gender differences significantly influence perceptions of empathy in hospital service quality.

Overall, males had a more negative perception of the overall service quality compared to females. The overall (p-value <0.05) confirms that gender significantly influences clients' perceptions of service quality across all five dimensions.

The findings suggest hospital administrators should consider gender-specific strategies to improve clients' perception of healthcare service quality.

4.7.2 Age

Statistical analysis using one-way ANOVA showed that there were statistically significant differences in gap scores for the responsiveness dimension based on age (p-value < 0.05), while for Tangibles, Reliability, Assurance and Empathy dimensions, the (p-value >0.05), implying that there are no statistically significant differences in gap scores for these dimensions based on age and that these aspects of service quality are perceived similarly across different age groups. This is demonstrated below in Table 4.16.

Within the responsiveness dimension, clients in the 45-59 age group had the most negative gap (-0.69 ± 1.04), indicating that they perceived a greater deviation of actual service from their expectations compared to younger (18-44) and older (>60) age groups.

Overall service quality was perceived negatively across all age groups, with the 45-59 age group reporting the larger negative gap. The >60 groups reported no overall gap, suggesting their expectations were met. The overall (p-value <0.05) indicated that age significantly influences clients' perceptions of service quality across all five dimensions.

Table 4.16: Age and Gap Scores

Age	Tangible	Reliability	Responsiveness	Assurance	Empathy	Overall
18-44	-0.27 ± 1.07	-0.43 ± 1.1	-0.4 ± 1.09	-0.42 ± 0.96	-0.37 ± 0.98	-0.38 ± 1.04
45-59	-0.44 ± 1.03	-0.53 ± 0.98	-0.69 ± 1.04	-0.58 ± 0.89	-0.47 ± 0.98	-0.54 ± 0.99
>60	0 ± 0.76	0 ± 1.41	0 ± 0.76	-0.12 ± 0.88	0.1 ± 1.37	0.0 ± 1.1
Prefer not to say	-0.56 ± 0.84	-0.78 ± 0.94	-0.58 ± 0.93	-0.31 ± 0.78	-0.51 ± 0.92	-0.43 ± 1.0
F- value	2.504	0.84	4.431	2.244	2.054	8.362
P-value	0.058	0.472	0.004	0.081	0.104	0

Primary data (2024)

Therefore, Halisi Family Hospital should pay particular attention to improving responsiveness, especially for clients in the 45-59 age group.

4.5.3 Education Level

Based on the analysis done using one-way ANOVA, there are statistically significant differences in the gap scores for the reliability dimension based on education (p-value < 0.05) while for the tangibles, responsiveness, assurance and empathy dimensions, the (p-value >0.05) implying that there are no statistically significant differences in gap scores for these dimensions based on education. This is demonstrated in Table 4.17 below.

Within the reliability dimension, postgraduate individuals reported the highest negative gap scores, indicating that they perceived a greater deviation of actual service from their expectations compared to those with secondary, certificate/diploma or bachelor's education.

Overall service quality was perceived negatively across all education levels, with the postgraduate group reporting the largest negative gap. The overall p-value (<0.05) indicated significant differences in overall service quality perceptions based on education level.

Table 4.17: Education Level and Gap Scores

Education	Tangible	Reliability	Responsiveness	Assurance	Empathy	Overall
Secondary	-0.19 ± 1.01	-0.31 ± 1.24	-0.36±1.23	-0.36 ±1.05	-0.38±1.09	-0.32 ± 1.12
Certificate/Diploma	-0.28 ± 1.03	-0.53 ± 0.89	-0.43 ± 0.96	-0.41± 0.79	-0.3 ± 1.0	-0.39 ± 0.94
Bachelors	0.39 ± 1.05	-0.44 ± 1.09	-0.55 ± 1.11	-0.56 ± 0.97	-0.44 ±1.01	-0.47 ± 1.05
Postgraduate	-0.46 ± 1.31	-0.82 ± 1.32	-0.71±1.23	-0.61±1.09	-0.53 ± 1.13	-0.63 ± 1.22
Prefer not to say	-0.17 ± 0.77	-0.32 ± 0.81	-0.26±0.86	-0.12±0.72	-0.32 ± 0.72	-0.43 ± 1.0
F-value	1.015	4.123	1.56	1.432	1.526	6.932
P-value	0.386	0.007	0.198	0.232	0.206	0

Primary data (2024)

Based on these results, Halisi Family Hospital should, therefore, pay particular attention to improving the perceived reliability of its services, especially for higher-educated clients, to enhance overall service quality.

4.5.4 Work Status

For the dimension's tangibles, reliability, and responsiveness, the differences in gap scores across work status groups were not statistically significant (p-value >0.05), as shown in Table 4.18 below. This implies that work status does not have an influence on the perception of healthcare service quality across these dimensions. For the assurance and empathy dimensions, the differences are statistically significant (p-value < 0.05), indicating that work status does influence the perceived quality of service in these dimensions.

Overall service quality was perceived negatively across all work status categories, with unemployed individuals reporting the largest negative gap. Retired individuals had the smallest negative gap. The overall p-value (<0.05) indicated significant differences in overall service quality perceptions based on work status.

Table 4.18: Work Status and Gap Scores

Work Status	Tangible	Reliability	Responsiveness	Assurance	Empathy	Overall
Employed	-0.34 ± 1.06	-0.47±1.08	-0.52 ± 1.08	-0.45±0.93	-0.42 ± 0.99	-0.44 ± 1.03
Self-Employed	-0.31±1.12	-0.45±1.02	-0.39 ± 1.14	-0.46±1.01	-0.28 ± 1.1	-0.38 ± 1.08
Unemployed	-0.41±1.07	-0.57±1.18	-0.69 ± 1.14	-0.73±0.81	-0.71±0.86	-0.62 ± 1.03
Retired	-0.12 ± 0.92	-0.25±1.2	-0.06 ± 0.96	-0.12±0.79	0.0 ± 1.1	-0.11 ± 1.0
Prefer not to say	-0.36 ± 0.74	-0.46±0.78	-0.47±0.76	-0.3±0.84	-0.35±0.71	-0.43 ± 1.0
F-value	0.328	0.315	1.711	2.673	4.497	7.503
P-value	0.859	0.868	0.146	0.031	0.001	0

Primary data (2024)

Based on these results, Halisi Family Hospital should prioritise improving the perceived assurance and empathy of its services, particularly for unemployed clients.

4.5.5 Mode of Payment for Healthcare Services

The results displayed in Table 4.19 below, show that there were statistically significant differences in gap scores across different modes of payment for the dimensions of tangibles (p-value = 0.000), responsiveness (p-value = 0.003), assurance (p-value = 0.000), and empathy (p-value = 0.000) where p-value was <0.05. This indicates that clients' perceptions of these aspects of service quality vary significantly depending on their payment method. However, in the reliability dimension, there was no statistically significant difference across different payment modes (p-value = 0.176); the p-value was greater than 0.05, implying that clients' perceptions of reliability are relatively consistent regardless of the payment method used.

In terms of mean gap scores, clients using only NHIF reported the most negative scores across all dimensions, with particularly low scores in responsiveness (-1.75±0.5) and tangibles (-1.5±1.0). This indicates that they perceived a greater deviation of actual service from their expectations compared to other payment methods. Conversely, clients paying out-of-pocket exhibited the least dissatisfaction with the tangibles (-0.14±1.35) and empathy (-0.13±1.26) dimensions.

Table 4.19: Mode of Payment and Gap Scores

Mode of Payment	Tangible	Reliability	Responsiveness	Assurance	Empathy	Overall
Private Health Insurance provided by my employer	-0.24 ± 0.96	-0.43 ± 1.02	-0.38 ± 1.03	-0.36 ± 0.9	-0.38 ± 0.93	-0.36 ± 0.97
NHIF Only	-1.5 ± 1.0	-1.4 ± 0.55	-1.75 ± 0.5	-1.5 ± 0.58	-1 ± 0.0	-1.41 ± 0.59
Private Health Insurance purchased by self	-0.36 ± 0.67	-0.49 ±1.04	-0.31 ± 0.79	-0.39 ± 0.80	-0.29 ± 1.06	-0.37 ± 0.9
Out of Pocket	-0.14 ± 1.35	-0.31 ± 1.32	-0.29 ±1.2	-0.38 ± 1.21	-0.13 ± 1.26	-0.25 ± 1.29
Prefer not to say	-0.65 ± 0.83	-0.58 ± 1.13	-0.75 ± 0.93	-0.9 ± 0.90	-0.65 ± 0.87	-0.7 ± 0.94
NHIF + Private Health Insurance purchased by self	-0.53 ± 0.69	-0.56 ± 0.85	-0.70 ± 0.83	-0.9 ± 0.90	-0.54 ± 0.67	-0.58 ± 0.74
NHIF + Private Health Insurance purchased by employer	-0.53 ± 0.69	-0.56 ± 0.85	-0.70 ± 0.83	-0.58 ± 0.59	-0.54 ± 0.67	-0.58 ± 0.74
F-value	3.870	1.369	2.489	4.541	4.403	11.569
P-value	0.000	0.176	0.003	0.000	0.000	0.000

Primary data (2024)

Overall service quality was perceived negatively across all payment modes, with the NHIF-only group reporting the largest negative gap (-1.41 ± 0.59). The out-of-pocket group had the smallest negative gap (-0.25 ± 1.29), indicating higher overall satisfaction compared to other payment modes. The overall p-value (<0.05) indicated significant differences in overall service quality perceptions based on the mode of payment.

These results suggest that Halisi Family Hospital administrators should prioritise improving service quality for NHIF users, especially in responsiveness and tangible dimensions, to reduce the gap between expectations and perceptions.

4.5.6 Income

From the results displayed in Table 4.20 below, there were no statistically significant differences across income levels for the dimensions of reliability (p-value = 0.864), responsiveness (p-value = 0.672), assurance (p-value = 0.701), and empathy (p-value = 0.301). This implies that clients' perceptions of these aspects of service quality are relatively consistent regardless of income level. However, in the tangibles dimension, there was a statistically significant difference in gap scores across different income levels (p-value = 0.004), which is less than 0.05.

Table 4.20: Income and gap scores of service quality dimension

Income	Tangible	Reliability	Responsiveness	Assurance	Empathy	Overall
49,999and below	-0.25 ± 1.02	-0.47 ± 0.95	-0.53 ± 1.02	-0.49 ± 0.83	-0.53 ± 0.69	-0.46 ± 0.91
50,000-99,999	-0.22 ± 0.95	-0.39 ± 0.96	-0.47 ± 0.99	-0.41 ± 0.84	-0.35 ± 0.89	-0.37 ± 0.93
100,000-149,999	-0.71 ± 1.32	-0.60 ±1.27	-0.62 ± 1.21	-0.57 ± 1.04	-0.45 ± 1.12	-0.58 ± 1.19
>150,000	-0.28 ± 1.27	-0.49 ± 1.33	-0.56 ±1.31	-0.38 ± 1.21	-0.29 ±1.27	-0.4 ± 1.28
Prefer not to say	-0.34 ± 0.96	-0.48 ± 1.03	-0.44 ± 1.04	-0.46 ± 0.92	-0.41 ± 1.01	-0.43 ± 1.0
F-value	3.821	0.321	0.588	0.548	1.202	3.957
P-value	0.004	0.864	0.672	0.701	0.301	0.003

Primary data (2024)

Within the tangibles dimension, individuals earning an average monthly income between 100,000 and 149,999 reported the highest negative gap scores, indicating that they perceived a greater deviation of actual service from their expectations compared to those earning below 100,000 and above 149,999.

Overall service quality was perceived negatively across all income categories, with the 100,000 - 149,999 group reporting the largest negative gap (-0.58 ± 1.19). The 50,000 - 99,999 group had the smallest negative gap (-0.37 ± 0.93), indicating higher overall satisfaction. The overall p-value (<0.05) indicated significant differences in overall service quality perceptions based on income.

As a result, Halisi Family Hospital administrators should focus on improving the tangible aspects of their services, particularly for clients with higher incomes, to bridge the gap between expectations and perceptions.

4.6 Relative Importance of SERVQUAL Dimensions

The respondents were asked to state the dimension that is most important to them. The results are displayed in Table 4.21 below.

Table 4.21: Relative Importance of SERVQUAL Dimensions

Relative Importance	Frequency	Percentage (%)
Reliability	68	39.1
Responsiveness	56	32.2
Assurance	24	13.8
Empathy	24	13.8
Tangibles	2	1.1
Total	**174**	**100.0**

Primary data (2024)

The majority of the respondents (68 respondents, 39.1%) chose reliability as the most important dimension. This suggests that a significant portion of clients prioritise the hospital's ability to perform services dependably and accurately. Only 2 respondents (1.1%) chose Tangibles, which include the physical facilities, equipment, and appearance of personnel, as the most important dimension. Physical aspects of the service are less critical to clients compared to other dimensions.